Step 3: Transform Data

Three common data transformations are scaling, attribute decompositions and attribute aggregations. This step is also referred to as feature engineering.

- Scaling: The preprocessed data may contain attributes with a mixture of scales for various quantities such as dollars, kilograms and sales volume. Many machine learning methods like data attributes to have the same scale such as between 0 and 1 for the smallest and largest value for a given feature. Consider any feature scaling you may need to perform.
- Decomposition: There may be features that represent a complex concept that may be more useful to a machine learning method when split into the constituent parts. An example is a date that may have day and time components that in turn could be split out further. Perhaps only the hour of day is relevant to the problem being solved. consider what feature decompositions you can perform.

- Aggregation: There may be features that can be aggregated into a single feature that would be more meaningful to the problem you are trying to solve. For example, there may be a data instances for each time a customer logged into a system that could be aggregated into a count for the number of logins allowing the additional instances to be discarded. Consider what type of feature aggregations could perform.

Summary

- Step 1: Data Selection: Consider what data is available, what data is missing and what data can be removed.
- Step 2: Data Preprocessing: Organize your selected data by formatting, cleaning and sampling from it.
- Step 3: Data Transformation: Transform preprocessed data ready for machine learning by engineering features using scaling, attribute decomposition and attribute aggregation.

Data preparation is a large subject that can involve a lot of iterations, exploration and analysis.

How to Improve Machine Learning Results

The three strategies you will learn about in this step are:

- Algorithm Tuning
- Ensembles
- Extreme Feature Engineering

- Algorithm Tuning where discovering the best models is treated like a search problem through model parameter space.
- Ensembles where the predictions made by multiple models are combined.
- Extreme Feature Engineering where the attribute decomposition and aggregation seen in data preparation is pushed to the limits.